DO NOW — RECALL

Lesson 077 · Linear Search: Code and Trace

Name: _____ Class: _____ Date: _____

1. Can linear search work with unsorted input?

2. What is not-found sentinel?

3. What does O(n) time mean?

4. Array a=[6,2,9,4]: set a[2]=a[0]+a[1]. Give updated array and total.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Yes.

2. Model answer

Distinct return such as −1.

3. Model answer

Work grows linearly with n.

4. Model answer

[6,2,8,4], sum20.

By the end, you can…

• Visit each array value in order; linear search does not require sorted data.

• Return the first matching index or a distinct not-found value such as −1.

• Worst-case O(n) time, O(1) extra storage; verify first, last, missing and empty inputs.

Vocabulary — know these terms

linear search — Sequentially compare elements.

sentinel — Special result indicating a condition.

index — Zero-based position.

NEW LEARNING

Read, annotate and explain

Understand

Visit each array value in order; linear search does not require sorted data.

Trace

Return the first matching index or a distinct not-found value such as −1.

Explain

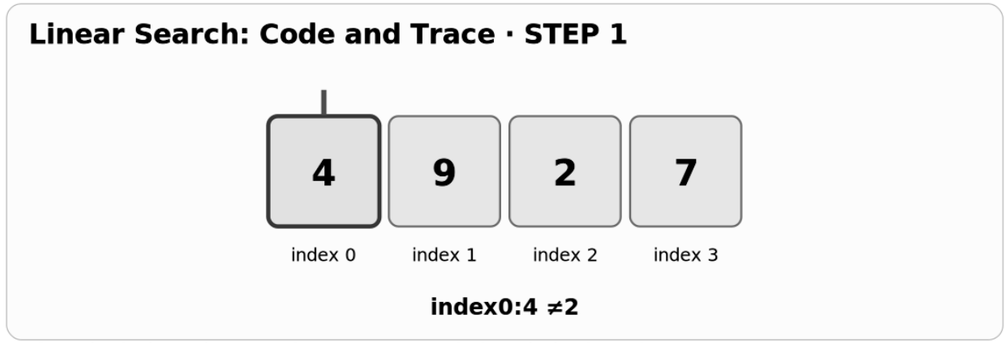
Worst-case O(n) time, O(1) extra storage; verify first, last, missing and empty inputs.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

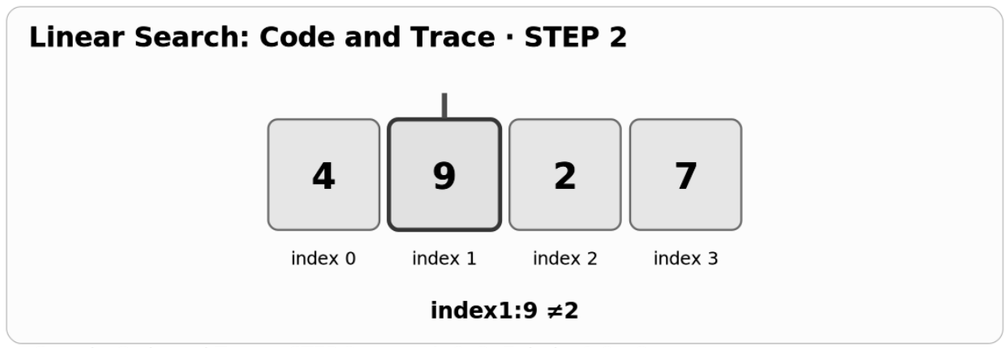
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Linear Search: Code and Trace · STEP 1; array 4, 9, 2, 7; index0:4 ≠2

New Learning



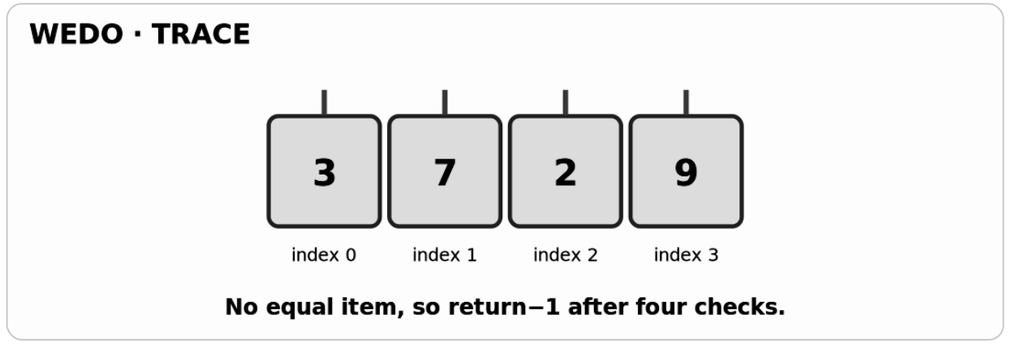
Linear Search: Code and Trace · STEP 2; array 4, 9, 2, 7; index1:9 ≠2

We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Find 8 in [3,7,2,9].

No equal item, so return−1 after four checks.



WEDO · TRACE; array 3, 7, 2, 9; No equal item, so return−1 after four checks.

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

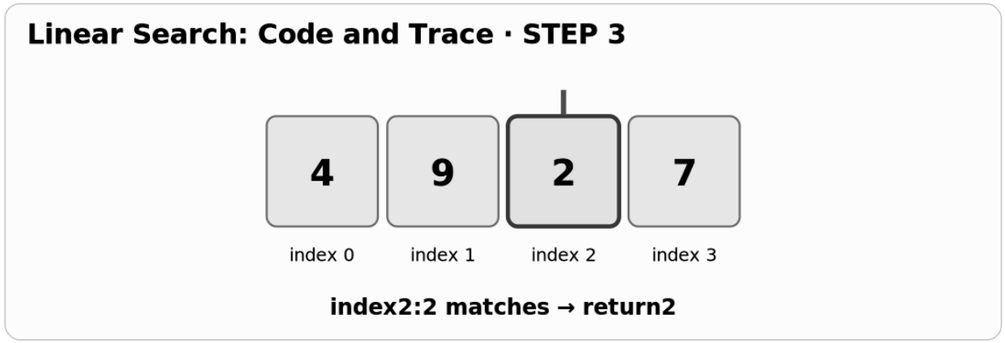
Guided 1 — CHECK

Check the working
No equal item, so return−1 after four checks.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Linear Search: Code and Trace · STEP 3; array 4, 9, 2, 7; index2:2 matches → return2

I DO — WORKED EXAMPLE

Follow the worked method

Find 6 in unsorted [1,5,6,3].

Check the working

Check indexes0,1,2, return2; stop before3.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

Linear search requires a pre-sorted array.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 3

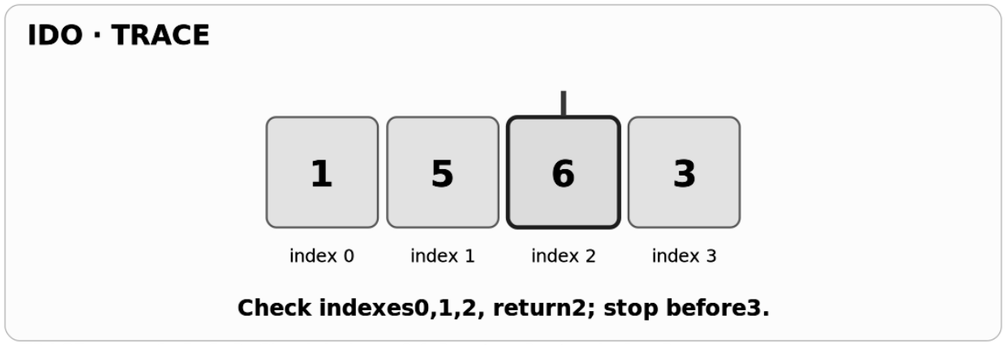
State linear search best/worst time, space and one edge test. [6]

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Find 6 in unsorted [1,5,6,3].

Check indexes0,1,2, return2; stop before3.



IDO · TRACE; array 1, 5, 6, 3; Check indexes0,1,2, return2; stop before3.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY

Find 8 in [3,7,2,9].
Source code / input:
No equal item, so return−1 after four checks.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

Trace 2 in [4,9,2,7], give checks and return. [6]

Question 2

Write C# linear search returning index/−1. [6]